Analysis of Kinetic Growth in Silicon Thermal Oxidation

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Abstract

Creating ultra-thin insulating layers on silicon wafers is crucial for semiconductor devices. This project explored the growth of silicon dioxide (SiO_2) using a combination of experiments and simulations. We investigated how factors like temperature affect the thickness and properties of the SiO_2 layer.

Introduction

- Silicon dioxide (SiO_2), also known as silicon oxide, plays a fundamental role in the miniaturization and performance of modern electronic devices. This incredibly thin insulating layer is an important feature in transistors depicted in **Figure 1** and forms the building blocks of integrated circuits (ICs). Unlike silicon, a semiconductor that allows controlled electrical current flow, SiO_2 is an insulator, meaning it effectively blocks current. This characteristic is crucial within transistors. By separating the conductive regions (source, drain, and gate) in a transistor with a thin layer of SiO_2, SiO_2 ensures proper current flow only when the gate voltage is applied, enabling efficient switching and signal amplification.
- Understanding the growth and properties of this essential insulating layer is critical for continued advancements in device miniaturization and performance. Thermal oxidation, a process where a silicon wafer is exposed to high temperatures in a controlled oxygen environment, allows for the formation of a thin SiO_2 layer directly on the silicon surface. The growth rate and properties of this layer are highly dependent on the oxidation temperature.
- The Deal-Grove model, established in a 1965 paper by Deal and Grove (1) , provides a fundamental framework for understanding the relationship between oxidation temperature, time, and the resulting thickness of the SiO_2 layer. This model allows researchers to predict growth rates and calculate activation energies, which are key parameters for optimizing the thermal oxidation process and tailoring the properties of the SiO_2 layer for specific device applications.
- This project investigated the thermal oxidation process of silicon wafers using a combination of experiments and modeling based on the Deal-Grove model. Our goal was to gain insights into the growth kinetics of SiO_2.

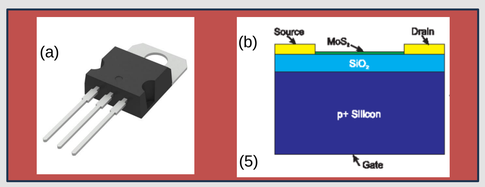


Figure 1 IRF520 MOSFET (a). Showing Silicon Oxide layer in a transistor (b)

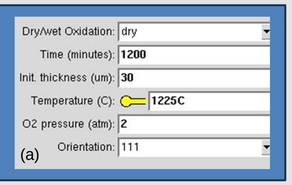
Methodology

➢ SiO_2 Modeling

- To gain insights into the growth kinetics of the (SiO_2) layer, we utilized the nanoHUB Process Lab: Oxidation (2) software for computational modeling. This software leverages the established Deal-Grove model (1) , which is a linear-parabolic model that incorporates critical parameters like oxidation temperature and time to predict the resulting thickness of the SiO_2 layer in **Figure 2**.

$$\frac{x_0^2}{B} + \frac{x_0}{B/A} = t + \tau \quad (1)$$

Figure 2 Oxidation simulation and inputs for one of the experiments (a)



➢ Modeling Results

- The Deal-Grove model (1) within the nanoHUB Process Lab software (2) , we simulated the thermal oxidation process to explore the relationship between oxidation temperature, time, and the resulting thickness of the SiO_2 layer. Simulations were conducted across a range of temperatures (e.g., 800°C - 1200°C) and varying oxidation times. The simulated data provided insights into the growth behavior of the SiO_2 layer, allowing us to analyze the impact of different oxidation conditions on the thickness in **Figure 3**.
- To further analyze the growth kinetics of the SiO_2 layer we utilized Arrhenius plots (3) . These plots depict the logarithm of the growth rate versus the reciprocal of the oxidation temperature. The slope of the Arrhenius plot has a direct relationship with the activation energy, a key parameter governing the rate-limiting step in the thermal oxidation process. By analyzing the slope of our simulated data plotted in Arrhenius form, we aimed to determine the activation energy for the growth of SiO_2 under the simulated conditions. This information provides valuable insights into the underlying mechanisms controlling the oxidation process in **Figure 3**.

$$\frac{B}{A} = C_2 \exp(-E_2/kT) \quad (4)$$

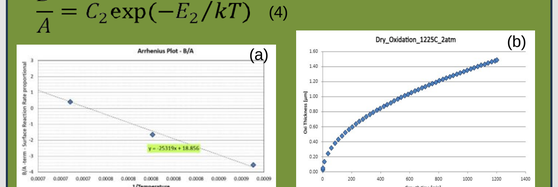


Figure 3 Arrhenius Plot representing Activation energy (a) Silicon Oxide thickness of a function of growth time 1225°C (b)

Oxidation Layer Characterization

➢ Experimental

- Our experiment utilized a thermal oxidation process to grow a silicon dioxide (SiO_2) layer on a silicon wafer. We utilized a bake oven for the thermal treatment. A 100 mm diameter silicon wafer, typically used in microfabrication, was subjected to high-temperature oxidation. To achieve the desired oxide layer growth, the wafer was placed in the bake oven (**Figure 4)** and exposed to a controlled temperature of 975 °C for an extended duration of 20 hours. The high temperature promotes the reaction between the silicon wafer and the ambient oxygen, leading to the formation of the SiO_2 layer on the silicon surface.

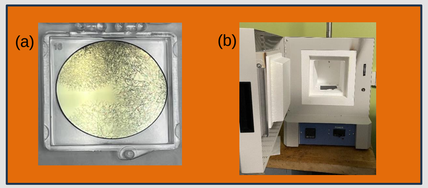


Figure 4 Silicon Wafer Results After Oxidation (a) Bake Oven in CGAM (b)

➢ Results and Characterization

- The newly oxidized wafer was examined using a Scanning Electron Microscope (SEM) to investigate the surface features of the grown SiO_2 layer and any potential structures.
- The SEM analysis provided information on the elemental composition of the SiO_2 film and presence of nanowires. This allowed us to confirm the presence of the desired oxide layer and identify any other elements that might be present on the surface shown in **Figure 5**.

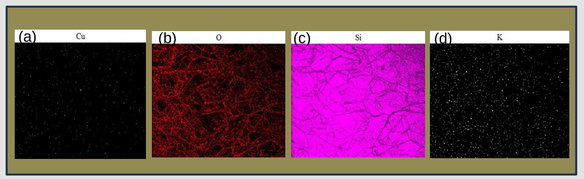


Figure 5 Silicon (c) and Oxygen (b) shown. Other elements present include Copper (a) and Potassium (d)

- Nanowires are essentially tiny structures with diameters in the nanometer range. These nanowires can have unique electrical and physical properties, making them potentially useful for various electronic applications (6) shown in **Figure 6**

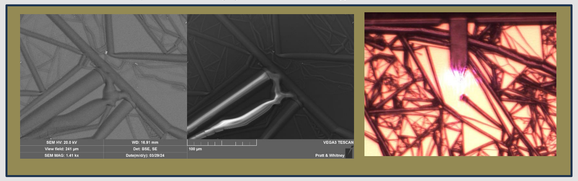


Figure 6 Nanowires shown at 100µm or 100000nm

Conclusion and Discussion

- Thermal oxidation successfully formed a silicon dioxide (SiO_2) layer on the silicon wafer. Scanning Electron Microscope (SEM) analysis revealed the presence of nanowire structures on the surface of the oxide layer.
- The formation of a high-quality SiO_2 layer is crucial for the development of miniaturized and efficient electronic devices. Furthermore, the unexpected discovery of nanowires on the oxide layer opens up exciting possibilities for future exploration of their electrical and physical properties for potential electronic applications.
- The simulated data obtained from the Deal-Grove model (1) provided valuable insights into the relationship between oxidation temperature and the growth rate of the SiO_2 layer. This information can be used to refine future oxidation processes for precise control over the oxide layer thickness
- For Future Optimization the data collected from the oxidation models and Arrhenius models can offer valuable insight on optimizing the thermal oxidation process. This can lead to better efficiency in the Semiconductor Fabrication Process.

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